

Supplementary material

What area and marital characteristics add to clinical need in predicting waits beyond 90 days for prostate cancer surgery or radiotherapy: a machine learning analysis of US SEER data, 2010 to 2022

How to read this supplement: - **Source of tables:** every table is copied by script from the generated analysis reports in the public repository (<https://github.com/tanmayhinge/seer-prostate>), so the tables match the analysis outputs exactly. - **Labels:** tables generated before revision 1 call the area and marital block "social position". - **Post-review analyses:** Supplementary Tables S7, S11 to S15, S21 and S22 come from analyses specified after an internal review (protocol amendment 1.9). S17 and S18 publish results computed under amendment 1.7, S35 re-tabulates sensitivity results computed under amendment 1.8, and S19a was tabulated after the second internal review from existing tables. - **Numbers of men:** in cross-tabulations they are rounded to the nearest 10, and statistics resting on 1 to 4 men are shown as <5. - **References:** numbered as in the main text.

Supplementary Box 1. Survival comparisons and immortal time bias (not estimated)

Comparing survival between treated and untreated men is exposed to immortal time bias, because treated men must survive until treatment starts. A simulation study of observational research found that time-fixed and exclusion methods overestimated a treatment's benefit, that a 1-year landmark method reduced the bias but did not remove it, and recommended the time-dependent method [29]. A suitable design would treat treatment as a time-dependent exposure, with competing risks of prostate cancer death and death from other causes, and use a 12-month landmark analysis only as a sensitivity analysis. This design is described, not estimated, in this study.

Supplementary Box 2. Candidate Australian analogues of the SEER variables (no Australian data analysed)

Published analyses of the Prostate Cancer Outcomes Registry in Tasmania used the measures below [24, 25]. They are candidate analogues, not equivalents: the geographic units, the construction of the measures and the health systems differ, and availability for any new study must be confirmed against the registry's data dictionary. - **Rurality:** the county Rural-Urban Continuum Code has a candidate analogue in Australian Statistical Geography Standard remoteness areas, assigned by residential postcode. - **Area income:** county median household income has a candidate analogue in the SEIFA Index of Relative Socio-Economic Advantage and Disadvantage, assigned by postcode. - **Insurance and sector:** SEER has no insurance variable. The closest Australian contrast studied is a public or private treating facility.

Data gaps reported in those analyses were no comorbidity data [24], patient-reported outcomes collected consistently only from 2018 [25], and no way to tell whether external beam radiotherapy was given in a public or private facility [25]. Referral, biopsy and multidisciplinary meeting dates are not assumed to be available.

Supplementary Table S1. Treated men by interval status

Men meeting cohort steps 1 to 6.

characteristic	level	Overall	Interval included	0 days	Missing interval
Men, n		355,581	330,827	7,753	17,001
Rurality	Metro, 1 million or more	210,372 (59.2%)	194,681 (58.8%)	4,388 (56.6%)	11,303 (66.5%)
Rurality	Metro, 250,000 to 1 million	77,524 (21.8%)	72,150 (21.8%)	1,737 (22.4%)	3,637 (21.4%)
Rurality	Metro, under 250,000	27,705 (7.8%)	26,092 (7.9%)	652 (8.4%)	961 (5.7%)
Rurality	Nonmetro, adjacent to metro	24,621 (6.9%)	23,360 (7.1%)	567 (7.3%)	694 (4.1%)
Rurality	Nonmetro, not adjacent to metro	15,188 (4.3%)	14,385 (4.3%)	404 (5.2%)	399 (2.3%)

characteristic	level	Overall	Interval included	0 days	Missing interval
Rurality	Unknown	171 (0.0%)	159 (0.0%)	5 (0.1%)	7 (0.0%)
Age at diagnosis	40-44 years	1,547 (0.4%)	1,480 (0.4%)	32 (0.4%)	35 (0.2%)
Age at diagnosis	45-49 years	7,297 (2.1%)	6,921 (2.1%)	128 (1.7%)	248 (1.5%)
Age at diagnosis	50-54 years	25,787 (7.3%)	24,293 (7.3%)	477 (6.2%)	1,017 (6.0%)
Age at diagnosis	55-59 years	52,043 (14.6%)	49,078 (14.8%)	894 (11.5%)	2,071 (12.2%)
Age at diagnosis	60-64 years	75,120 (21.1%)	70,459 (21.3%)	1,368 (17.6%)	3,293 (19.4%)
Age at diagnosis	65-69 years	89,835 (25.3%)	83,831 (25.3%)	1,769 (22.8%)	4,235 (24.9%)
Age at diagnosis	70-74 years	61,150 (17.2%)	56,371 (17.0%)	1,473 (19.0%)	3,306 (19.4%)
Age at diagnosis	75-79 years	31,004 (8.7%)	28,114 (8.5%)	947 (12.2%)	1,943 (11.4%)
Age at diagnosis	80-84 years	9,834 (2.8%)	8,636 (2.6%)	491 (6.3%)	707 (4.2%)
Age at diagnosis	85-89 years	1,792 (0.5%)	1,516 (0.5%)	147 (1.9%)	129 (0.8%)
Age at diagnosis	90+ years	172 (0.0%)	128 (0.0%)	27 (0.3%)	17 (0.1%)
Year of diagnosis	2010 to 2014	138,812 (39.0%)	126,503 (38.2%)	2,967 (38.3%)	9,342 (54.9%)
Year of diagnosis	2015 to 2019	131,356 (36.9%)	122,936 (37.2%)	3,056 (39.4%)	5,364 (31.6%)
Year of diagnosis	2020 to 2022	85,413 (24.0%)	81,388 (24.6%)	1,730 (22.3%)	2,295 (13.5%)
Marital status	Married (including common law)	249,351 (70.1%)	232,525 (70.3%)	5,321 (68.6%)	11,505 (67.7%)
Marital status	Single (never married)	39,507 (11.1%)	36,703 (11.1%)	881 (11.4%)	1,923 (11.3%)
Marital status	Divorced	23,631 (6.6%)	22,023 (6.7%)	565 (7.3%)	1,043 (6.1%)
Marital status	Separated	2,853 (0.8%)	2,630 (0.8%)	67 (0.9%)	156 (0.9%)
Marital status	Widowed	10,885 (3.1%)	9,905 (3.0%)	341 (4.4%)	639 (3.8%)
Marital status	Unmarried or Domestic Partner	1,447 (0.4%)	1,369 (0.4%)	24 (0.3%)	54 (0.3%)
Marital status	Unknown	27,907 (7.8%)	25,672 (7.8%)	554 (7.1%)	1,681 (9.9%)
County median household income (quartile of 16 bands)	Q1 (lowest)	27,790 (7.8%)	26,436 (8.0%)	<5	<5
County median household income (quartile of 16 bands)	Q2	86,090 (24.2%)	79,804 (24.1%)	1,886 (24.3%)	4,400 (25.9%)
County median household income (quartile of 16 bands)	Q3	125,843 (35.4%)	116,211 (35.1%)	2,747 (35.4%)	6,885 (40.5%)
County median household income (quartile of 16 bands)	Q4 (highest)	115,822 (32.6%)	108,348 (32.8%)	2,323 (30.0%)	5,151 (30.3%)
County median household income (quartile of 16 bands)	Unknown	36 (0.0%)	28 (0.0%)	<5	<5
Risk group	low	60,793 (17.1%)	57,495 (17.4%)	863 (11.1%)	2,435 (14.3%)
Risk group	intermediate	134,413 (37.8%)	127,622 (38.6%)	1,766 (22.8%)	5,025 (29.6%)
Risk group	high	141,384 (39.8%)	130,893 (39.6%)	3,434 (44.3%)	7,057 (41.5%)
Risk group	unknown	18,991 (5.3%)	14,817 (4.5%)	1,690 (21.8%)	2,484 (14.6%)
Summary stage	Localised	276,161 (77.7%)	256,475 (77.5%)	6,025 (77.7%)	13,661 (80.4%)
Summary stage	Regional, direct extension	66,083 (18.6%)	61,984 (18.7%)	1,301 (16.8%)	2,798 (16.5%)
Summary stage	Regional, lymph nodes	13,337 (3.8%)	12,368 (3.7%)	427 (5.5%)	542 (3.2%)
Clinical Gleason score	6 or less	87,024 (24.5%)	81,630 (24.7%)	1,406 (18.1%)	3,988 (23.5%)
Clinical Gleason score	7	181,097 (50.9%)	171,345 (51.8%)	2,688 (34.7%)	7,064 (41.6%)
Clinical Gleason score	8 to 10	79,786 (22.4%)	73,447 (22.2%)	2,164 (27.9%)	4,175 (24.6%)
Clinical Gleason score	Unknown	7,674 (2.2%)	4,405 (1.3%)	1,495 (19.3%)	1,774 (10.4%)
PSA (ng/ml)	below 10	236,726 (66.6%)	223,700 (67.6%)	3,954 (51.0%)	9,072 (53.4%)
PSA (ng/ml)	10 to 20	62,849 (17.7%)	58,930 (17.8%)	984 (12.7%)	2,935 (17.3%)
PSA (ng/ml)	above 20	31,563 (8.9%)	28,789 (8.7%)	944 (12.2%)	1,830 (10.8%)
PSA (ng/ml)	Unknown	24,443 (6.9%)	19,408 (5.9%)	1,871 (24.1%)	3,164 (18.6%)
First-course treatment	Radical prostatectomy	170,715 (48.0%)	159,580 (48.2%)	3,746 (48.3%)	7,389 (43.5%)

characteristic	level	Overall	Interval included	0 days	Missing interval
First-course treatment	Radiotherapy	171,287 (48.2%)	158,473 (47.9%)	3,672 (47.4%)	9,142 (53.8%)
First-course treatment	Both	13,579 (3.8%)	12,774 (3.9%)	335 (4.3%)	470 (2.8%)

Supplementary Tables S2 to S6. Crude waiting beyond 90 days

Medians and 90th percentiles treat top-coded intervals as 731 days.

Supplementary Table S2. By risk group

risk group	men	% waited over 90 days	median days	90th percentile days
low	57,495	47.9	88.0	210.0
intermediate	127,622	42.8	82.0	176.0
high	130,893	32.8	70.0	150.0
unknown	14,817	42.1	79.0	212.0

Supplementary Table S3. By risk group and county rurality

risk group	rurality	men	% waited over 90 days	median days	90th percentile days
low	Metro, 1 million or more	34,260	50.7	91.0	217.0
low	Metro, 250,000 to 1 million	11,930	47.0	86.0	211.0
low	Metro, under 250,000	4,780	42.3	82.0	198.2
low	Nonmetro, adjacent to metro	3,990	39.0	77.0	187.0
low	Nonmetro, not adjacent to metro	2,510	39.0	76.0	177.0
low	Unknown	20	34.8	81.0	122.8
intermediate	Metro, 1 million or more	75,190	45.5	84.0	182.0
intermediate	Metro, 250,000 to 1 million	28,140	41.1	79.0	172.0
intermediate	Metro, under 250,000	10,070	36.7	75.0	165.0
intermediate	Nonmetro, adjacent to metro	8,830	36.1	74.0	161.0
intermediate	Nonmetro, not adjacent to metro	5,360	36.1	71.0	160.0
intermediate	Unknown	40	42.5	71.0	200.5
high	Metro, 1 million or more	75,760	35.7	73.0	157.0
high	Metro, 250,000 to 1 million	29,320	30.0	67.0	142.0
high	Metro, under 250,000	10,130	27.9	65.0	140.0
high	Nonmetro, adjacent to metro	9,590	27.7	63.0	134.5
high	Nonmetro, not adjacent to metro	6,010	26.0	62.0	133.0
high	Unknown	90	28.4	61.5	141.5
unknown	Metro, 1 million or more	9,460	44.2	82.0	224.0
unknown	Metro, 250,000 to 1 million	2,770	42.2	80.0	203.0
unknown	Metro, under 250,000	1,120	35.0	70.0	179.0
unknown	Nonmetro, adjacent to metro	950	36.5	72.0	174.2
unknown	Nonmetro, not adjacent to metro	510	29.6	63.5	162.0
unknown	Unknown	10	<5	77.0	212.7

Supplementary Table S4. By risk group and county income quartile

risk group	county income quartile	men	% waited over 90 days	median days	90th percentile days
low	Q1 (lowest)	5,420	37.6	76.0	182.0
low	Q2	15,540	44.1	83.0	196.0
low	Q3	19,460	50.4	91.0	216.0
low	Q4 (highest)	17,070	51.9	93.0	225.0
low	Unknown	<5	<5	<5	<5
intermediate	Q1 (lowest)	9,930	33.5	70.0	155.0
intermediate	Q2	30,100	39.3	77.0	168.0
intermediate	Q3	44,260	44.9	84.0	184.0
intermediate	Q4 (highest)	43,330	45.2	84.0	179.0
intermediate	Unknown	10	<5	71.0	372.2
high	Q1 (lowest)	9,920	26.7	63.0	134.8
high	Q2	30,420	30.5	67.0	145.0
high	Q3	46,890	35.0	71.0	156.0

risk group	county income quartile	men	% waited over 90 days	median days	90th percentile days
high	Q4 (highest)	43,650	33.4	70.0	151.0
high	Unknown	10	38.5	82.0	146.2
unknown	Q1 (lowest)	1,160	31.4	65.0	175.7
unknown	Q2	3,740	39.3	76.0	207.0
unknown	Q3	5,610	44.6	83.0	222.5
unknown	Q4 (highest)	4,300	44.4	83.0	210.0
unknown	Unknown	<5	<5	<5	<5

Supplementary Table S5. By risk group and marital status

risk group	marital status	men	% waited over 90 days	median days	90th percentile days
low	Married (including common law)	41,490	47.0	87.0	204.0
low	Single (never married)	5,690	51.9	93.0	226.0
low	Divorced	3,300	50.0	90.0	220.2
low	Separated	410	52.1	92.0	231.6
low	Widowed	1,200	42.5	83.0	196.0
low	Unmarried or Domestic Partner	160	59.9	106.5	245.0
low	Unknown	5,250	49.8	90.0	243.0
intermediate	Married (including common law)	89,430	41.3	80.0	169.2
intermediate	Single (never married)	14,160	49.4	90.0	197.0
intermediate	Divorced	8,840	47.4	87.0	190.0
intermediate	Separated	1,040	49.3	90.0	196.0
intermediate	Widowed	4,030	40.4	78.0	176.0
intermediate	Unmarried or Domestic Partner	570	49.6	90.0	194.5
intermediate	Unknown	9,560	42.6	81.0	189.0
high	Married (including common law)	92,380	30.9	68.0	143.0
high	Single (never married)	15,390	40.0	77.0	172.0
high	Divorced	9,110	36.6	74.0	161.0
high	Separated	1,110	41.4	80.0	177.0
high	Widowed	4,320	30.9	63.0	148.0
high	Unmarried or Domestic Partner	600	40.3	77.0	166.6
high	Unknown	7,980	36.6	73.0	168.0
unknown	Married (including common law)	9,230	40.2	77.0	197.4
unknown	Single (never married)	1,460	47.3	87.0	222.2
unknown	Divorced	770	46.4	83.0	223.0
unknown	Separated	80	48.1	87.0	311.0
unknown	Widowed	360	40.0	77.0	208.6
unknown	Unmarried or Domestic Partner	40	50.0	90.0	255.0
unknown	Unknown	2,890	44.6	82.0	253.0

Supplementary Table S6. By year of diagnosis

year of diagnosis	men	% waited over 90 days	median days	90th percentile days
2010	29,740	36.0	73.0	166.0
2011	29,350	35.8	73.0	161.0
2012	23,830	34.0	71.0	160.0
2013	22,530	33.5	71.0	159.0
2014	21,060	34.4	72.0	160.0
2015	22,530	36.2	74.0	162.0
2016	23,300	38.1	76.0	168.0
2017	24,120	39.3	77.0	164.0
2018	25,730	41.4	79.0	181.0
2019	27,270	43.5	82.0	187.0
2020	23,870	41.3	79.0	178.0
2021	28,750	46.3	85.0	188.0
2022	28,780	52.3	93.0	199.0

Supplementary Table S7. Long and top-coded intervals (post-review)

variable	group	% of men waiting more than 365 days	% of men top-coded (731 days or more)
risk group	intermediate	1.3	0.25
risk group	high	0.8	0.17
risk group	low	3.0	0.80
risk group	unknown	4.2	1.57
rurality	Metro, 1 million or more	1.7	0.40
rurality	Metro, 250,000 to 1 million	1.4	0.35
rurality	Nonmetro, adjacent to metro	1.1	0.23
rurality	Nonmetro, not adjacent to metro	0.9	0.22
rurality	Metro, under 250,000	1.5	0.41
rurality	Unknown	<5	<5

Supplementary Table S8. Model performance at every step

AUC: 0.5 is chance. Calibration intercept: 0 is ideal. Calibration slope: 1 is ideal.

stratum	model	step	log-loss skill %	Brier skill %	AUC	calibration intercept	calibration slope
all men (pooled)	penalised logistic regression	0	0.00	0.00	0.558	-0.000	1.001
all men (pooled)	penalised logistic regression	1	3.01	3.89	0.628	-0.000	1.002
all men (pooled)	penalised logistic regression	2	3.63	4.66	0.638	-0.000	1.000
all men (pooled)	penalised logistic regression	3	3.67	4.71	0.638	-0.000	1.000
all men (pooled)	LightGBM	0	0.00	0.00	0.561	-0.000	0.997
all men (pooled)	LightGBM	1	3.18	4.09	0.633	-0.000	1.004
all men (pooled)	LightGBM	2	4.17	5.34	0.648	-0.000	1.030
all men (pooled)	LightGBM	3	4.46	5.70	0.653	-0.000	1.041
low risk	penalised logistic regression	0	0.00	0.00	0.566	0.000	1.005
low risk	penalised logistic regression	1	0.14	0.19	0.573	-0.000	0.998
low risk	penalised logistic regression	2	0.76	1.03	0.592	0.000	0.988
low risk	penalised logistic regression	3	0.97	1.31	0.597	0.000	0.989
low risk	LightGBM	0	0.00	0.00	0.570	-0.000	0.989
low risk	LightGBM	1	0.02	0.04	0.577	-0.000	0.883
low risk	LightGBM	2	0.99	1.35	0.604	-0.000	0.937
low risk	LightGBM	3	1.35	1.84	0.611	-0.000	0.957
intermediate risk	penalised logistic regression	0	0.00	0.00	0.581	0.000	1.003
intermediate risk	penalised logistic regression	1	0.49	0.66	0.594	-0.000	1.006
intermediate risk	penalised logistic regression	2	1.06	1.40	0.605	-0.000	1.001
intermediate risk	penalised logistic regression	3	1.09	1.45	0.606	-0.000	1.001
intermediate risk	LightGBM	0	0.00	0.00	0.582	0.000	0.993
intermediate risk	LightGBM	1	0.55	0.74	0.596	0.000	0.966
intermediate risk	LightGBM	2	1.48	1.96	0.615	-0.000	1.008
intermediate risk	LightGBM	3	1.66	2.20	0.619	-0.000	1.022
high risk	penalised logistic regression	0	0.00	0.00	0.564	-0.000	1.004
high risk	penalised logistic regression	1	3.53	4.41	0.641	-0.000	1.002
high risk	penalised logistic regression	2	4.39	5.42	0.654	0.000	1.001
high risk	penalised logistic regression	3	4.63	5.71	0.657	0.000	1.001
high risk	LightGBM	0	0.00	0.00	0.565	-0.000	0.993
high risk	LightGBM	1	3.72	4.63	0.645	-0.000	0.992
high risk	LightGBM	2	4.90	6.04	0.663	-0.000	1.023
high risk	LightGBM	3	5.28	6.49	0.668	-0.000	1.033
unknown risk	penalised logistic regression	0	0.00	0.00	0.584	-0.000	1.019
unknown risk	penalised logistic regression	1	0.65	0.89	0.601	-0.000	1.022
unknown risk	penalised logistic regression	2	1.07	1.43	0.610	0.000	0.982
unknown risk	penalised logistic regression	3	1.41	1.86	0.616	0.000	0.980
unknown risk	LightGBM	0	0.00	0.00	0.585	-0.000	0.946
unknown risk	LightGBM	1	0.59	0.83	0.601	-0.001	0.850
unknown risk	LightGBM	2	1.96	2.64	0.629	-0.000	0.882
unknown risk	LightGBM	3	2.67	3.54	0.640	-0.000	0.901

Supplementary Table S9. Skill lost when one variable is removed from step 2

Rurality and county income are strongly correlated, so removing one lets the other partly stand in for it.

stratum	model	variable removed	skill lost when removed (points)
all men (pooled)	penalised logistic regression	marital status	0.22
all men (pooled)	penalised logistic regression	rurality	0.23
all men (pooled)	penalised logistic regression	county income	0.00
all men (pooled)	LightGBM	marital status	0.27
all men (pooled)	LightGBM	rurality	0.23
all men (pooled)	LightGBM	county income	0.26
low risk	penalised logistic regression	marital status	0.05
low risk	penalised logistic regression	rurality	0.13
low risk	penalised logistic regression	county income	0.09
low risk	LightGBM	marital status	0.05
low risk	LightGBM	rurality	0.10
low risk	LightGBM	county income	0.38
intermediate risk	penalised logistic regression	marital status	0.18
intermediate risk	penalised logistic regression	rurality	0.19
intermediate risk	penalised logistic regression	county income	0.01
intermediate risk	LightGBM	marital status	0.21
intermediate risk	LightGBM	rurality	0.22
intermediate risk	LightGBM	county income	0.29
high risk	penalised logistic regression	marital status	0.42
high risk	penalised logistic regression	rurality	0.37
high risk	penalised logistic regression	county income	0.02
high risk	LightGBM	marital status	0.45
high risk	LightGBM	rurality	0.36
high risk	LightGBM	county income	0.26
unknown risk	penalised logistic regression	marital status	0.16
unknown risk	penalised logistic regression	rurality	0.12
unknown risk	penalised logistic regression	county income	-0.03
unknown risk	LightGBM	marital status	0.41
unknown risk	LightGBM	rurality	0.16
unknown risk	LightGBM	county income	0.59

Supplementary Table S10. Tuned hyperparameters (primary analysis, tuned at step 3)

stratum	model	max_iter	C	seconds	learning_rate	n_estimators	num_leaves	min_child_samples
all men (pooled)	penalised logistic regression	2000	0.01	1.9	n/a	n/a	n/a	n/a
all men (pooled)	LightGBM	n/a	n/a	41.6	0.05	200	15	200
low risk	penalised logistic regression	2000	0.01	1.3	n/a	n/a	n/a	n/a
low risk	LightGBM	n/a	n/a	40.7	0.05	200	15	200
intermediate risk	penalised logistic regression	2000	0.01	1.4	n/a	n/a	n/a	n/a
intermediate risk	LightGBM	n/a	n/a	40.5	0.05	200	15	200
high risk	penalised logistic regression	2000	0.01	1.9	n/a	n/a	n/a	n/a
high risk	LightGBM	n/a	n/a	42.4	0.05	200	15	200
unknown risk	penalised logistic regression	2000	0.01	0.4	n/a	n/a	n/a	n/a
unknown risk	LightGBM	n/a	n/a	30.7	0.05	200	15	200

Supplementary Tables S11 to S15. Post-review robustness analyses of the primary estimand

Supplementary Table S11. Area and marital increment with per-step tuning and repeated fold assignment

Intervals come from the first fold seed and do not include model-fitting variability; the seed range shows how much the estimate moves across 10 fold assignments.

stratum	model	social position added, primary (step 3 tuning)	social position added, per-step tuning, first fold seed	mean (minimum to maximum) over 10 fold seeds	interval with county income band clusters
all men (pooled)	penalised logistic regression	0.62 (0.39 to 0.87)	0.62 (0.39 to 0.87)	0.62 (0.61 to 0.62)	0.62 (0.38 to 0.90)
all men (pooled)	LightGBM	0.99 (0.73 to 1.30)	0.99 (0.73 to 1.30)	0.99 (0.98 to 0.99)	0.99 (0.72 to 1.35)
low risk	penalised logistic regression	0.62 (0.32 to 0.94)	0.62 (0.32 to 0.94)	0.62 (0.59 to 0.63)	0.62 (0.28 to 1.07)
low risk	LightGBM	0.96 (0.67 to 1.28)	0.96 (0.67 to 1.28)	0.98 (0.95 to 1.01)	0.96 (0.64 to 1.36)
intermediate risk	penalised logistic regression	0.56 (0.35 to 0.83)	0.56 (0.34 to 0.83)	0.56 (0.54 to 0.56)	0.56 (0.33 to 0.89)
intermediate risk	LightGBM	0.93 (0.65 to 1.27)	0.93 (0.65 to 1.27)	0.93 (0.91 to 0.97)	0.93 (0.63 to 1.31)
high risk	penalised logistic regression	0.86 (0.60 to 1.15)	0.86 (0.60 to 1.15)	0.85 (0.85 to 0.86)	0.86 (0.58 to 1.15)
high risk	LightGBM	1.18 (0.90 to 1.56)	1.18 (0.90 to 1.54)	1.18 (1.16 to 1.20)	1.18 (0.87 to 1.55)
unknown risk	penalised logistic regression	0.42 (0.08 to 0.80)	0.40 (0.07 to 0.80)	0.41 (0.32 to 0.48)	0.40 (0.07 to 0.78)
unknown risk	LightGBM	1.37 (0.80 to 2.19)	1.37 (0.80 to 2.19)	1.29 (1.22 to 1.43)	1.37 (0.78 to 2.09)

Supplementary Table S12. Clinical model comparison under per-step tuning

stratum	clinical need added, logistic (seed mean)	clinical need added, LightGBM (seed mean)	LightGBM clinical model at least as good	step 2 skill, logistic (seed mean)	step 2 skill, LightGBM (seed mean)
all men (pooled)	3.01	3.18	yes	3.63	4.17
low risk	0.14	0.05	no	0.76	1.03
intermediate risk	0.49	0.53	yes	1.05	1.47
high risk	3.53	3.72	yes	4.38	4.90
unknown risk	0.64	0.55	no	1.05	1.84

Supplementary Table S13. Hyperparameters chosen at each step

stratum	model	step	max_iter	C	learning_rate	n_estimators	num_leaves	min_child_samples
all men (pooled)	penalised logistic regression	0	2000	0.01	n/a	n/a	n/a	n/a
all men (pooled)	penalised logistic regression	1	2000	0.01	n/a	n/a	n/a	n/a
all men (pooled)	penalised logistic regression	2	2000	0.01	n/a	n/a	n/a	n/a
all men (pooled)	LightGBM	0	n/a	n/a	0.05	200	15	50
all men (pooled)	LightGBM	1	n/a	n/a	0.05	200	15	50
all men (pooled)	LightGBM	2	n/a	n/a	0.05	200	15	200
low risk	penalised logistic regression	0	2000	0.1	n/a	n/a	n/a	n/a
low risk	penalised logistic regression	1	2000	0.01	n/a	n/a	n/a	n/a
low risk	penalised logistic regression	2	2000	0.01	n/a	n/a	n/a	n/a
low risk	LightGBM	0	n/a	n/a	0.05	200	15	50
low risk	LightGBM	1	n/a	n/a	0.05	200	15	200
low risk	LightGBM	2	n/a	n/a	0.05	200	15	200
intermediate risk	penalised logistic regression	0	2000	0.1	n/a	n/a	n/a	n/a
intermediate risk	penalised logistic regression	1	2000	0.1	n/a	n/a	n/a	n/a
intermediate risk	penalised logistic regression	2	2000	0.01	n/a	n/a	n/a	n/a
intermediate risk	LightGBM	0	n/a	n/a	0.05	200	15	50
intermediate risk	LightGBM	1	n/a	n/a	0.05	200	15	200
intermediate risk	LightGBM	2	n/a	n/a	0.05	200	15	200
high risk	penalised logistic regression	0	2000	0.1	n/a	n/a	n/a	n/a
high risk	penalised logistic regression	1	2000	0.1	n/a	n/a	n/a	n/a
high risk	penalised logistic regression	2	2000	0.01	n/a	n/a	n/a	n/a
high risk	LightGBM	0	n/a	n/a	0.05	200	15	50
high risk	LightGBM	1	n/a	n/a	0.05	200	15	50
high risk	LightGBM	2	n/a	n/a	0.05	200	15	200
unknown risk	penalised logistic regression	0	2000	0.1	n/a	n/a	n/a	n/a
unknown risk	penalised logistic regression	1	2000	0.1	n/a	n/a	n/a	n/a
unknown risk	penalised logistic regression	2	2000	0.01	n/a	n/a	n/a	n/a
unknown risk	LightGBM	0	n/a	n/a	0.05	200	15	50
unknown risk	LightGBM	1	n/a	n/a	0.05	200	15	200

stratum	model	step	max_iter	C	learning_rate	n_estimators	num_leaves	min_child_samples
unknown risk	LightGBM	2	n/a	n/a	0.05	200	15	200

Supplementary Table S14. Bootstrap clusters by stratum

stratum	clustering	clusters	smallest cluster (men)	median cluster (men)	largest cluster (men)	largest cluster, % of men
all men (pooled)	rurality by income cells	79	23	1,870	28,195	8.5
all men (pooled)	county income band only	17	28	22,703	36,147	10.9
low risk	rurality by income cells	78	<5	310	5,288	9.2
low risk	county income band only	17	<5	3,342	6,807	11.8
intermediate risk	rurality by income cells	79	7	659	11,724	9.2
intermediate risk	county income band only	17	7	8,879	14,291	11.2
high risk	rurality by income cells	79	12	717	11,366	8.7
high risk	county income band only	17	13	9,352	14,824	11.3
unknown risk	rurality by income cells	77	<5	70	1,226	8.3
unknown risk	county income band only	17	<5	967	1,644	11.1

Supplementary Table S15. Stage-free clinical block and year as categories

analysis	stratum	model	comparison	estimate (95% interval)	primary
stage-free clinical block	all men (pooled)	penalised logistic regression	clinical need (step 0 to 1)	2.97 (2.64 to 3.34)	3.01 (2.69 to 3.38)
stage-free clinical block	all men (pooled)	penalised logistic regression	social position (step 1 to 2)	0.62 (0.39 to 0.87)	0.62 (0.39 to 0.87)
stage-free clinical block	all men (pooled)	LightGBM	clinical need (step 0 to 1)	3.13 (2.80 to 3.48)	3.18 (2.85 to 3.54)
stage-free clinical block	all men (pooled)	LightGBM	social position (step 1 to 2)	1.02 (0.75 to 1.33)	0.99 (0.73 to 1.30)
stage-free clinical block	low risk	penalised logistic regression	clinical need (step 0 to 1)	0.14 (0.08 to 0.20)	0.14 (0.08 to 0.20)
stage-free clinical block	low risk	penalised logistic regression	social position (step 1 to 2)	0.62 (0.32 to 0.94)	0.62 (0.32 to 0.94)
stage-free clinical block	low risk	LightGBM	clinical need (step 0 to 1)	0.02 (-0.09 to 0.13)	0.02 (-0.09 to 0.13)
stage-free clinical block	low risk	LightGBM	social position (step 1 to 2)	0.96 (0.67 to 1.28)	0.96 (0.67 to 1.28)
stage-free clinical block	intermediate risk	penalised logistic regression	clinical need (step 0 to 1)	0.49 (0.41 to 0.57)	0.49 (0.41 to 0.56)
stage-free clinical block	intermediate risk	penalised logistic regression	social position (step 1 to 2)	0.56 (0.34 to 0.83)	0.56 (0.35 to 0.83)
stage-free clinical block	intermediate risk	LightGBM	clinical need (step 0 to 1)	0.55 (0.44 to 0.65)	0.55 (0.44 to 0.65)
stage-free clinical block	intermediate risk	LightGBM	social position (step 1 to 2)	0.93 (0.65 to 1.27)	0.93 (0.65 to 1.27)
stage-free clinical block	high risk	penalised logistic regression	clinical need (step 0 to 1)	3.42 (3.07 to 3.76)	3.53 (3.16 to 3.86)
stage-free clinical block	high risk	penalised logistic regression	social position (step 1 to 2)	0.85 (0.59 to 1.14)	0.86 (0.60 to 1.15)
stage-free clinical block	high risk	LightGBM	clinical need (step 0 to 1)	3.64 (3.27 to 3.99)	3.72 (3.34 to 4.06)
stage-free clinical block	high risk	LightGBM	social position (step 1 to 2)	1.19 (0.91 to 1.57)	1.18 (0.90 to 1.56)
stage-free clinical block	unknown risk	penalised logistic regression	clinical need (step 0 to 1)	0.66 (0.41 to 0.92)	0.65 (0.41 to 0.89)
stage-free clinical block	unknown risk	penalised logistic regression	social position (step 1 to 2)	0.40 (0.06 to 0.80)	0.42 (0.08 to 0.80)
stage-free clinical block	unknown risk	LightGBM	clinical need (step 0 to 1)	0.59 (0.27 to 0.95)	0.59 (0.27 to 0.95)
stage-free clinical block	unknown risk	LightGBM	social position (step 1 to 2)	1.37 (0.80 to 2.19)	1.37 (0.80 to 2.19)
year as categories	all men (pooled)	penalised logistic regression	clinical need (step 0 to 1)	2.98 (2.66 to 3.34)	3.01 (2.69 to 3.38)
year as categories	all men (pooled)	penalised logistic regression	social position (step 1 to 2)	0.62 (0.39 to 0.86)	0.62 (0.39 to 0.87)
year as categories	low risk	penalised logistic regression	clinical need (step 0 to 1)	0.15 (0.09 to 0.21)	0.14 (0.08 to 0.20)
year as categories	low risk	penalised logistic regression	social position (step 1 to 2)	0.60 (0.31 to 0.92)	0.62 (0.32 to 0.94)
year as categories	intermediate risk	penalised logistic regression	clinical need (step 0 to 1)	0.50 (0.42 to 0.58)	0.49 (0.41 to 0.56)
year as categories	intermediate risk	penalised logistic regression	social position (step 1 to 2)	0.56 (0.34 to 0.83)	0.56 (0.35 to 0.83)
year as categories	high risk	penalised logistic regression	clinical need (step 0 to 1)	3.48 (3.12 to 3.82)	3.53 (3.16 to 3.86)
year as categories	high risk	penalised logistic regression	social position (step 1 to 2)	0.87 (0.61 to 1.16)	0.86 (0.60 to 1.15)
year as categories	unknown risk	penalised logistic regression	clinical need (step 0 to 1)	0.65 (0.41 to 0.90)	0.65 (0.41 to 0.89)
year as categories	unknown risk	penalised logistic regression	social position (step 1 to 2)	0.47 (0.11 to 0.89)	0.42 (0.08 to 0.80)

Supplementary Table S16. Standardised percentage waiting more than 90 days by profile, all men

profile	penalised logistic regression	LightGBM
as observed	39.7	39.7
reference profile	40.8	41.3
area: Metro, 1 million or more, typical county income (\$90,000 - \$94,999)	40.6	41.1
area: Metro, 250,000 to 1 million, typical county income (\$80,000 - \$84,999)	36.3	36.9
area: Metro, under 250,000, typical county income (\$65,000 - \$69,999)	32.8	32.4
area: Nonmetro, adjacent to metro, typical county income (\$55,000 - \$59,999)	31.8	31.9

profile	penalised logistic regression	LightGBM
area: Nonmetro, not adjacent to metro, typical county income (\$55,000 - \$59,999)	31.4	32.4
marital status: Married (including common law)	40.8	41.3
marital status: Single (never married)	47.5	47.2
marital status: Divorced	46.9	46.7
marital status: Separated	49.4	46.9
marital status: Widowed	42.7	44.0
marital status: Unmarried or Domestic Partner	46.7	45.9
marital status: Unknown	45.6	45.7

Supplementary Tables S17 and S18. One-at-a-time rurality and income profiles (not interpreted)

These profiles change rurality or county income alone, holding the other at the reference profile, which creates combinations rarely observed in the data. The two model types conflict, which is why joint area profiles are reported instead (amendment 1.7).

Supplementary Table S17. Contrasts (percentage points)

stratum	contrast	penalised logistic regression	LightGBM	agreement
all men (pooled)	rurality only: Nonmetro, not adjacent to metro minus Metro, 1 million or more	-8.7	1.7	opposite directions, at least one model 3 points or more
all men (pooled)	county income only: Q1 (lowest) minus Q4 (highest)	-0.9	-8.7	same direction, only one model 3 points or more
low risk	rurality only: Nonmetro, not adjacent to metro minus Metro, 1 million or more	-7.3	-4.9	both models 3 points or more, same direction
low risk	county income only: Q1 (lowest) minus Q4 (highest)	-7.0	-14.8	both models 3 points or more, same direction
intermediate risk	rurality only: Nonmetro, not adjacent to metro minus Metro, 1 million or more	-7.7	-2.5	same direction, only one model 3 points or more
intermediate risk	county income only: Q1 (lowest) minus Q4 (highest)	-2.0	-10.9	same direction, only one model 3 points or more
high risk	rurality only: Nonmetro, not adjacent to metro minus Metro, 1 million or more	-9.9	-3.3	both models 3 points or more, same direction
high risk	county income only: Q1 (lowest) minus Q4 (highest)	3.4	-4.8	opposite directions, at least one model 3 points or more
unknown risk	rurality only: Nonmetro, not adjacent to metro minus Metro, 1 million or more	-11.4	-6.2	both models 3 points or more, same direction
unknown risk	county income only: Q1 (lowest) minus Q4 (highest)	-1.5	-9.6	same direction, only one model 3 points or more

Supplementary Table S18. Standardised percentages

stratum	model	profile	standardised %
all men (pooled)	penalised logistic regression	rurality only: Metro, 1 million or more (county income held at reference)	40.8
all men (pooled)	penalised logistic regression	rurality only: Metro, 250,000 to 1 million (county income held at reference)	36.6
all men (pooled)	penalised logistic regression	rurality only: Metro, under 250,000 (county income held at reference)	33.3
all men (pooled)	penalised logistic regression	rurality only: Nonmetro, adjacent to metro (county income held at reference)	32.4
all men (pooled)	penalised logistic regression	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	32.1
all men (pooled)	penalised logistic regression	county income only: Q1 (lowest) (rurality held at reference)	39.9
all men (pooled)	penalised logistic regression	county income only: Q2 (rurality held at reference)	40.2
all men (pooled)	penalised logistic regression	county income only: Q3 (rurality held at reference)	40.5
all men (pooled)	penalised logistic regression	county income only: Q4 (highest) (rurality held at reference)	40.8
all men (pooled)	LightGBM	rurality only: Metro, 1 million or more (county income held at reference)	41.3
all men (pooled)	LightGBM	rurality only: Metro, 250,000 to 1 million (county income held at reference)	37.4
all men (pooled)	LightGBM	rurality only: Metro, under 250,000 (county income held at reference)	37.4
all men (pooled)	LightGBM	rurality only: Nonmetro, adjacent to metro (county income held at reference)	38.0
all men (pooled)	LightGBM	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	43.0
all men (pooled)	LightGBM	county income only: Q1 (lowest) (rurality held at reference)	32.6
all men (pooled)	LightGBM	county income only: Q2 (rurality held at reference)	34.6
all men (pooled)	LightGBM	county income only: Q3 (rurality held at reference)	40.4
all men (pooled)	LightGBM	county income only: Q4 (highest) (rurality held at reference)	41.3
low risk	penalised logistic regression	rurality only: Metro, 1 million or more (county income held at reference)	51.3

stratum	model	profile	standardised %
low risk	penalised logistic regression	rurality only: Metro, 250,000 to 1 million (county income held at reference)	48.9
low risk	penalised logistic regression	rurality only: Metro, under 250,000 (county income held at reference)	46.1
low risk	penalised logistic regression	rurality only: Nonmetro, adjacent to metro (county income held at reference)	42.8
low risk	penalised logistic regression	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	44.0
low risk	penalised logistic regression	county income only: Q1 (lowest) (rurality held at reference)	44.6
low risk	penalised logistic regression	county income only: Q2 (rurality held at reference)	46.9
low risk	penalised logistic regression	county income only: Q3 (rurality held at reference)	49.3
low risk	penalised logistic regression	county income only: Q4 (highest) (rurality held at reference)	51.6
low risk	LightGBM	rurality only: Metro, 1 million or more (county income held at reference)	52.6
low risk	LightGBM	rurality only: Metro, 250,000 to 1 million (county income held at reference)	50.0
low risk	LightGBM	rurality only: Metro, under 250,000 (county income held at reference)	50.0
low risk	LightGBM	rurality only: Nonmetro, adjacent to metro (county income held at reference)	48.7
low risk	LightGBM	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	47.7
low risk	LightGBM	county income only: Q1 (lowest) (rurality held at reference)	37.8
low risk	LightGBM	county income only: Q2 (rurality held at reference)	41.5
low risk	LightGBM	county income only: Q3 (rurality held at reference)	47.7
low risk	LightGBM	county income only: Q4 (highest) (rurality held at reference)	52.6
intermediate risk	penalised logistic regression	rurality only: Metro, 1 million or more (county income held at reference)	44.2
intermediate risk	penalised logistic regression	rurality only: Metro, 250,000 to 1 million (county income held at reference)	40.2
intermediate risk	penalised logistic regression	rurality only: Metro, under 250,000 (county income held at reference)	36.7
intermediate risk	penalised logistic regression	rurality only: Nonmetro, adjacent to metro (county income held at reference)	36.0
intermediate risk	penalised logistic regression	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	36.6
intermediate risk	penalised logistic regression	county income only: Q1 (lowest) (rurality held at reference)	42.3
intermediate risk	penalised logistic regression	county income only: Q2 (rurality held at reference)	43.0
intermediate risk	penalised logistic regression	county income only: Q3 (rurality held at reference)	43.6
intermediate risk	penalised logistic regression	county income only: Q4 (highest) (rurality held at reference)	44.3
intermediate risk	LightGBM	rurality only: Metro, 1 million or more (county income held at reference)	44.4
intermediate risk	LightGBM	rurality only: Metro, 250,000 to 1 million (county income held at reference)	40.9
intermediate risk	LightGBM	rurality only: Metro, under 250,000 (county income held at reference)	43.0
intermediate risk	LightGBM	rurality only: Nonmetro, adjacent to metro (county income held at reference)	40.9
intermediate risk	LightGBM	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	41.9
intermediate risk	LightGBM	county income only: Q1 (lowest) (rurality held at reference)	33.5
intermediate risk	LightGBM	county income only: Q2 (rurality held at reference)	37.0
intermediate risk	LightGBM	county income only: Q3 (rurality held at reference)	43.5
intermediate risk	LightGBM	county income only: Q4 (highest) (rurality held at reference)	44.4
high risk	penalised logistic regression	rurality only: Metro, 1 million or more (county income held at reference)	32.7
high risk	penalised logistic regression	rurality only: Metro, 250,000 to 1 million (county income held at reference)	27.2
high risk	penalised logistic regression	rurality only: Metro, under 250,000 (county income held at reference)	24.5
high risk	penalised logistic regression	rurality only: Nonmetro, adjacent to metro (county income held at reference)	24.2
high risk	penalised logistic regression	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	22.9
high risk	penalised logistic regression	county income only: Q1 (lowest) (rurality held at reference)	35.9
high risk	penalised logistic regression	county income only: Q2 (rurality held at reference)	34.8
high risk	penalised logistic regression	county income only: Q3 (rurality held at reference)	33.7
high risk	penalised logistic regression	county income only: Q4 (highest) (rurality held at reference)	32.6
high risk	LightGBM	rurality only: Metro, 1 million or more (county income held at reference)	32.9
high risk	LightGBM	rurality only: Metro, 250,000 to 1 million (county income held at reference)	27.8
high risk	LightGBM	rurality only: Metro, under 250,000 (county income held at reference)	28.0
high risk	LightGBM	rurality only: Nonmetro, adjacent to metro (county income held at reference)	29.0
high risk	LightGBM	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	29.6
high risk	LightGBM	county income only: Q1 (lowest) (rurality held at reference)	28.1
high risk	LightGBM	county income only: Q2 (rurality held at reference)	29.2
high risk	LightGBM	county income only: Q3 (rurality held at reference)	33.7
high risk	LightGBM	county income only: Q4 (highest) (rurality held at reference)	32.9
unknown risk	penalised logistic regression	rurality only: Metro, 1 million or more (county income held at reference)	42.3
unknown risk	penalised logistic regression	rurality only: Metro, 250,000 to 1 million (county income held at reference)	41.2
unknown risk	penalised logistic regression	rurality only: Metro, under 250,000 (county income held at reference)	34.4
unknown risk	penalised logistic regression	rurality only: Nonmetro, adjacent to metro (county income held at reference)	36.2
unknown risk	penalised logistic regression	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	30.8
unknown risk	penalised logistic regression	county income only: Q1 (lowest) (rurality held at reference)	40.9
unknown risk	penalised logistic regression	county income only: Q2 (rurality held at reference)	41.4

stratum	model	profile	standardised %
unknown risk	penalised logistic regression	county income only: Q3 (rurality held at reference)	41.9
unknown risk	penalised logistic regression	county income only: Q4 (highest) (rurality held at reference)	42.3
unknown risk	LightGBM	rurality only: Metro, 1 million or more (county income held at reference)	42.1
unknown risk	LightGBM	rurality only: Metro, 250,000 to 1 million (county income held at reference)	42.0
unknown risk	LightGBM	rurality only: Metro, under 250,000 (county income held at reference)	35.0
unknown risk	LightGBM	rurality only: Nonmetro, adjacent to metro (county income held at reference)	41.5
unknown risk	LightGBM	rurality only: Nonmetro, not adjacent to metro (county income held at reference)	36.0
unknown risk	LightGBM	county income only: Q1 (lowest) (rurality held at reference)	32.5
unknown risk	LightGBM	county income only: Q2 (rurality held at reference)	32.7
unknown risk	LightGBM	county income only: Q3 (rurality held at reference)	42.5
unknown risk	LightGBM	county income only: Q4 (highest) (rurality held at reference)	42.1

Supplementary Table S19. Crude excess waiting days beyond 90 days per 1,000 men

stratum	variable	group	men	excess days per 1,000 men (crude)
all men (pooled)	rurality	Metro, 1 million or more	194,680	30,597
all men (pooled)	rurality	Metro, 250,000 to 1 million	72,150	26,040
all men (pooled)	rurality	Metro, under 250,000	26,090	24,387
all men (pooled)	rurality	Nonmetro, adjacent to metro	23,360	21,678
all men (pooled)	rurality	Nonmetro, not adjacent to metro	14,390	20,014
all men (pooled)	rurality	Unknown	160	24,258
all men (pooled)	county income quartile	Q1 (lowest)	26,440	22,045
all men (pooled)	county income quartile	Q2	79,800	25,568
all men (pooled)	county income quartile	Q3	116,210	30,250
all men (pooled)	county income quartile	Q4 (highest)	108,350	28,889
all men (pooled)	county income quartile	Unknown	30	45,571
high risk	rurality	Metro, 1 million or more	75,760	21,912
high risk	rurality	Metro, 250,000 to 1 million	29,320	17,331
high risk	rurality	Metro, under 250,000	10,130	16,770
high risk	rurality	Nonmetro, adjacent to metro	9,590	15,316
high risk	rurality	Nonmetro, not adjacent to metro	6,010	13,553
high risk	rurality	Unknown	90	11,216
high risk	county income quartile	Q1 (lowest)	9,920	15,461
high risk	county income quartile	Q2	30,420	17,899
high risk	county income quartile	Q3	46,890	21,596
high risk	county income quartile	Q4 (highest)	43,650	19,625
high risk	county income quartile	Unknown	10	18,462

Supplementary Table S19a. Share of crude excess waiting days from top-coded intervals

Tabulated after the second internal review from existing tables; all men.

rurality	% of crude excess waiting days from top-coded intervals
Metro, 1 million or more	8.4
Metro, 250,000 to 1 million	8.6
Metro, under 250,000	10.9
Nonmetro, adjacent to metro	6.8
Nonmetro, not adjacent to metro	7.1
Unknown	<5

Supplementary Table S20. Erreygers concentration index by risk group

Crude within each stratum; it does not separate income from rurality.

stratum	Erreygers index (95% interval)
all men (pooled)	0.062 (0.039 to 0.083)
low risk	0.103 (0.068 to 0.134)

stratum	Erreygers index (95% interval)
intermediate risk	0.075 (0.051 to 0.096)
high risk	0.041 (0.017 to 0.065)
unknown risk	0.077 (0.037 to 0.128)

Supplementary Tables S21 and S22. Concentration index by diagnosis period (post-review)

Supplementary Table S21. Erreygers index by period

stratum	period	Erreygers index (95% interval)
all men (pooled)	2010 to 2014	0.059 (0.025 to 0.092)
all men (pooled)	2015 to 2019	0.037 (0.011 to 0.065)
all men (pooled)	2020 to 2022	0.027 (0.003 to 0.061)
high risk	2010 to 2014	0.028 (-0.011 to 0.062)
high risk	2015 to 2019	0.020 (-0.004 to 0.049)
high risk	2020 to 2022	0.015 (-0.013 to 0.056)
intermediate risk	2010 to 2014	0.058 (0.023 to 0.099)
intermediate risk	2015 to 2019	0.048 (0.019 to 0.080)
intermediate risk	2020 to 2022	0.039 (0.015 to 0.067)
low risk	2010 to 2014	0.092 (0.053 to 0.136)
low risk	2015 to 2019	0.084 (0.052 to 0.121)
low risk	2020 to 2022	0.052 (0.017 to 0.092)
unknown risk	2010 to 2014	0.060 (0.001 to 0.130)
unknown risk	2015 to 2019	0.023 (-0.041 to 0.100)
unknown risk	2020 to 2022	0.049 (-0.005 to 0.101)

Supplementary Table S22. County income quartile distribution by period

period	Q1 (lowest)	Q2	Q3	Q4 (highest)	Unknown
2010 to 2014	9.7	30.4	33.7	26.2	0.0
2015 to 2019	7.9	20.9	37.1	34.1	0.0
2020 to 2022	5.4	19.3	34.3	41.0	<5

Supplementary Tables S23 to S26. Bounds for men without a recorded interval

The lower bound assumes every man without a recorded interval waited 90 days or less; the upper bound assumes every such man waited longer. These bounds do not address selection into recorded treatment.

Supplementary Table S23. By risk group

risk group	men	% over 90 days (recorded men)	lower bound %	upper bound %	% with no recorded interval
low	59,930	47.9	46.0	50.0	4.1
intermediate	132,650	42.8	41.2	44.9	3.8
high	137,950	32.8	31.1	36.2	5.1
unknown	17,300	42.1	36.1	50.5	14.4

Supplementary Table S24. By rurality

rurality	men	% over 90 days (recorded men)	lower bound %	upper bound %	% with no recorded interval
Metro, 1 million or more	205,980	42.5	40.2	45.7	5.5
Metro, 250,000 to 1 million	75,790	37.6	35.8	40.6	4.8
Metro, under 250,000	27,050	34.2	33.0	36.6	3.6
Nonmetro, adjacent to metro	24,050	33.2	32.2	35.1	2.9
Nonmetro, not adjacent to metro	14,780	32.2	31.3	34.0	2.7
Unknown	170	33.3	31.9	36.1	4.2

Supplementary Table S25. By county income quartile

county income quartile	men	% over 90 days (recorded men)	lower bound %	upper bound %	% with no recorded interval
Q1 (lowest)	27,000	31.7	31.0	33.1	2.1
Q2	84,200	36.9	35.0	40.2	5.2
Q3	123,100	41.8	39.5	45.1	5.6
Q4 (highest)	113,500	41.4	39.6	44.1	4.5
Unknown	30	39.3	32.4	50.0	17.6

Supplementary Table S26. By marital status

marital status	men	% over 90 days (recorded men)	lower bound %	upper bound %	% with no recorded interval
Married (including common law)	244,030	38.1	36.3	41.0	4.7
Single (never married)	38,630	45.7	43.5	48.4	5.0
Divorced	23,070	43.3	41.3	45.8	4.5
Separated	2,790	46.3	43.8	49.4	5.6
Widowed	10,540	36.5	34.3	40.3	6.1
Unmarried or Domestic Partner	1,420	46.7	45.0	48.8	3.8
Unknown	27,350	42.4	39.8	46.0	6.1

Supplementary Table S27. Skill added in predicting a missing interval

comparison	skill added, percentage points (95% interval)
clinical need (step 0 to 1)	4.99 (4.51 to 5.42)
social position (step 1 to 2)	0.53 (0.07 to 1.02)

Supplementary Table S28. Inverse probability weighted percentages

variable	group	men	% over 90 days (unweighted)	% over 90 days (weighted)
rurality	Metro, 1 million or more	194,680	42.5	42.3
rurality	Metro, 250,000 to 1 million	72,150	37.6	37.4
rurality	Metro, under 250,000	26,090	34.2	34.1
rurality	Nonmetro, adjacent to metro	23,360	33.2	33.1
rurality	Nonmetro, not adjacent to metro	14,390	32.2	32.0
rurality	Unknown	160	33.3	33.5
county income quartile	Q1 (lowest)	26,440	31.7	31.5
county income quartile	Q2	79,800	36.9	36.8
county income quartile	Q3	116,210	41.8	41.6
county income quartile	Q4 (highest)	108,350	41.4	41.3
county income quartile	Unknown	30	39.3	39.6
marital status	Married (including common law)	232,530	38.1	38.0
marital status	Single (never married)	36,700	45.7	45.5
marital status	Divorced	22,020	43.3	43.1
marital status	Separated	2,630	46.3	46.2
marital status	Widowed	9,910	36.5	36.3
marital status	Unmarried or Domestic Partner	1,370	46.7	46.6
marital status	Unknown	25,670	42.4	42.4

Supplementary Tables S29 to S35. Sensitivity analyses

Supplementary Table S29. Percentage waiting beyond the threshold, by scenario

scenario	threshold (days)	men	% over threshold, all men	% over threshold, low risk	% over threshold, intermediate risk	% over threshold, high risk	% over threshold, unknown risk
primary analysis	90	330,830	39.7	47.9	42.8	32.8	42.1
threshold of 60 days	60	330,830	66.6	75.0	70.0	59.5	66.7
threshold of 120 days	120	330,830	23.0	30.3	24.8	17.5	26.8
threshold of 180 days	180	330,830	9.1	13.9	9.5	6.1	13.3

scenario	threshold (days)	men	% over threshold, all men	% over threshold, low risk	% over threshold, intermediate risk	% over threshold, high risk	% over threshold, unknown risk
risk groups from Gleason score and PSA only	90	330,830	39.7	47.7	42.6	28.2	41.7
radical prostatectomy without radiotherapy only	90	159,580	41.1	45.8	42.3	38.4	38.2
excluding men diagnosed in 2020	90	306,960	39.6	47.4	42.5	32.9	41.7
including men diagnosed in 2023	90	358,540	40.8	48.4	44.2	34.0	43.5
including intervals of 0 days	90	338,580	38.8	47.2	42.2	32.0	37.8
one primary cancer only (strict first primary)	90	296,120	40.1	48.3	43.2	33.2	42.7
excluding prostatectomy not otherwise specified	90	330,180	39.7	47.9	42.8	32.8	42.2

Supplementary Table S30. Area and marital increment by scenario

scenario	model	all men (pooled)	low risk	intermediate risk	high risk	unknown risk
primary analysis	penalised logistic regression	0.62 (0.39 to 0.87)	0.62 (0.32 to 0.94)	0.56 (0.35 to 0.83)	0.86 (0.60 to 1.15)	0.42 (0.08 to 0.80)
primary analysis	LightGBM	0.99 (0.73 to 1.30)	0.96 (0.67 to 1.28)	0.93 (0.65 to 1.27)	1.18 (0.90 to 1.56)	1.37 (0.80 to 2.19)
threshold of 60 days	penalised logistic regression	0.54 (0.31 to 0.79)	0.82 (0.38 to 1.27)	0.53 (0.28 to 0.82)	0.61 (0.39 to 0.86)	0.41 (0.06 to 0.86)
threshold of 60 days	LightGBM	0.94 (0.73 to 1.18)	1.22 (0.86 to 1.67)	0.88 (0.66 to 1.16)	0.96 (0.75 to 1.21)	1.56 (0.90 to 2.36)
threshold of 120 days	penalised logistic regression	0.68 (0.45 to 0.93)	0.60 (0.31 to 0.90)	0.58 (0.37 to 0.82)	1.02 (0.75 to 1.33)	0.52 (0.10 to 1.00)
threshold of 120 days	LightGBM	1.06 (0.79 to 1.38)	0.97 (0.69 to 1.23)	0.90 (0.58 to 1.25)	1.39 (1.10 to 1.77)	1.47 (0.71 to 2.40)
threshold of 180 days	penalised logistic regression	0.66 (0.41 to 0.92)	0.56 (0.32 to 0.82)	0.58 (0.33 to 0.83)	0.96 (0.64 to 1.31)	0.58 (0.00 to 1.19)
threshold of 180 days	LightGBM	1.00 (0.69 to 1.35)	0.84 (0.56 to 1.10)	0.93 (0.60 to 1.33)	1.40 (1.05 to 1.81)	1.33 (0.14 to 2.85)
risk groups from Gleason score and PSA only	penalised logistic regression	0.62 (0.39 to 0.87)	0.67 (0.34 to 1.01)	0.58 (0.38 to 0.82)	0.94 (0.65 to 1.28)	0.51 (0.16 to 0.87)
risk groups from Gleason score and PSA only	LightGBM	0.99 (0.73 to 1.30)	1.02 (0.71 to 1.32)	0.92 (0.67 to 1.23)	1.34 (1.00 to 1.78)	1.21 (0.59 to 1.92)
radical prostatectomy without radiotherapy only	penalised logistic regression	0.66 (0.48 to 0.88)	0.71 (0.47 to 1.07)	0.59 (0.37 to 0.83)	0.77 (0.56 to 0.99)	0.62 (0.19 to 1.01)
radical prostatectomy without radiotherapy only	LightGBM	1.00 (0.79 to 1.26)	0.88 (0.60 to 1.19)	0.90 (0.65 to 1.20)	1.07 (0.85 to 1.36)	0.74 (-0.13 to 1.42)
excluding men diagnosed in 2020	penalised logistic regression	0.64 (0.40 to 0.91)	0.59 (0.28 to 0.97)	0.58 (0.33 to 0.86)	0.89 (0.62 to 1.20)	0.43 (0.03 to 0.83)
excluding men diagnosed in 2020	LightGBM	1.03 (0.75 to 1.35)	0.95 (0.67 to 1.27)	0.98 (0.66 to 1.35)	1.20 (0.91 to 1.58)	1.33 (0.61 to 2.21)
including men diagnosed in 2023	penalised logistic regression	0.56 (0.35 to 0.80)	0.57 (0.27 to 0.89)	0.51 (0.31 to 0.75)	0.76 (0.51 to 1.04)	0.39 (0.08 to 0.75)
including men diagnosed in 2023	LightGBM	0.94 (0.69 to 1.22)	0.98 (0.72 to 1.28)	0.87 (0.61 to 1.19)	1.10 (0.86 to 1.46)	1.17 (0.54 to 1.95)
including intervals of 0 days	penalised logistic regression	0.61 (0.38 to 0.86)	0.63 (0.32 to 0.95)	0.55 (0.33 to 0.81)	0.83 (0.57 to 1.12)	0.52 (0.16 to 0.91)
including intervals of 0 days	LightGBM	0.97 (0.70 to 1.27)	1.04 (0.77 to 1.35)	0.92 (0.63 to 1.26)	1.14 (0.87 to 1.50)	1.31 (0.65 to 2.16)
one primary cancer only (strict first primary)	penalised logistic regression	0.61 (0.38 to 0.85)	0.63 (0.33 to 0.95)	0.54 (0.32 to 0.79)	0.85 (0.59 to 1.12)	0.40 (-0.04 to 0.79)
one primary cancer only (strict first primary)	LightGBM	0.98 (0.73 to 1.27)	0.98 (0.71 to 1.27)	0.90 (0.62 to 1.24)	1.22 (0.93 to 1.58)	1.39 (0.70 to 2.26)
excluding prostatectomy not otherwise specified	penalised logistic regression	0.62 (0.39 to 0.87)	0.61 (0.30 to 0.94)	0.54 (0.32 to 0.80)	0.86 (0.59 to 1.15)	0.52 (0.15 to 0.93)
excluding prostatectomy not otherwise specified	LightGBM	0.98 (0.72 to 1.28)	1.03 (0.76 to 1.32)	0.93 (0.65 to 1.28)	1.19 (0.92 to 1.55)	1.37 (0.74 to 2.13)

Supplementary Table S31. Standardised area contrast by scenario, all men

scenario	penalised logistic regression	LightGBM	agreement
primary analysis	-9.2	-8.8	both models 3 points or more, same direction
threshold of 60 days	-10.6	-10.3	both models 3 points or more, same direction

scenario	penalised logistic regression	LightGBM	agreement
threshold of 120 days	-6.9	-7.1	both models 3 points or more, same direction
threshold of 180 days	-3.0	-3.2	same direction, only one model 3 points or more
risk groups from Gleason score and PSA only	-9.2	-8.8	both models 3 points or more, same direction
radical prostatectomy without radiotherapy only	-8.9	-7.8	both models 3 points or more, same direction
excluding men diagnosed in 2020	-9.4	-8.9	both models 3 points or more, same direction
including men diagnosed in 2023	-8.6	-8.3	both models 3 points or more, same direction
including intervals of 0 days	-9.2	-9.2	both models 3 points or more, same direction
one primary cancer only (strict first primary)	-9.1	-9.3	both models 3 points or more, same direction
excluding prostatectomy not otherwise specified	-9.2	-9.1	both models 3 points or more, same direction

Supplementary Table S32. Standardised marital status contrast by scenario, all men

scenario	penalised logistic regression	LightGBM	agreement
primary analysis	6.7	5.9	both models 3 points or more, same direction
threshold of 60 days	3.7	3.2	both models 3 points or more, same direction
threshold of 120 days	6.0	5.0	both models 3 points or more, same direction
threshold of 180 days	3.4	3.1	both models 3 points or more, same direction
risk groups from Gleason score and PSA only	6.7	5.9	both models 3 points or more, same direction
radical prostatectomy without radiotherapy only	7.1	6.9	both models 3 points or more, same direction
excluding men diagnosed in 2020	6.8	5.9	both models 3 points or more, same direction
including men diagnosed in 2023	6.4	5.8	both models 3 points or more, same direction
including intervals of 0 days	6.5	5.8	both models 3 points or more, same direction
one primary cancer only (strict first primary)	6.6	5.6	both models 3 points or more, same direction
excluding prostatectomy not otherwise specified	6.7	5.9	both models 3 points or more, same direction

Supplementary Table S33. All area and marital features as observed minus the reference profile, by scenario, all men

scenario	penalised logistic regression	LightGBM	agreement
primary analysis	-1.1	-1.6	both models under 3 points
threshold of 60 days	-1.7	-2.8	both models under 3 points
threshold of 120 days	-0.6	-0.6	both models under 3 points
threshold of 180 days	0.1	0.3	both models under 3 points
risk groups from Gleason score and PSA only	-1.1	-1.6	both models under 3 points
radical prostatectomy without radiotherapy only	-1.5	-1.1	both models under 3 points
excluding men diagnosed in 2020	-1.1	-1.6	both models under 3 points
including men diagnosed in 2023	-1.0	-1.6	both models under 3 points
including intervals of 0 days	-1.1	-1.6	both models under 3 points
one primary cancer only (strict first primary)	-1.1	-1.6	both models under 3 points
excluding prostatectomy not otherwise specified	-1.1	-1.6	both models under 3 points

Supplementary Table S34. Penalised logistic regression with a wider C grid (post hoc)

stratum	C chosen (wider grid)	at an edge of the wider grid	logistic step 2 skill, original grid	logistic step 2 skill, wider grid	LightGBM step 2 skill	social position added, original grid	social position added, wider grid
all men (pooled)	0.01	no	3.63	3.63	4.17	0.62 (0.39 to 0.87)	0.62 (0.39 to 0.87)
low risk	0.01	no	0.76	0.76	0.99	0.62 (0.32 to 0.94)	0.62 (0.32 to 0.94)
intermediate risk	0.01	no	1.06	1.06	1.48	0.56 (0.35 to 0.83)	0.56 (0.35 to 0.83)
high risk	0.01	no	4.39	4.39	4.90	0.86 (0.60 to 1.15)	0.86 (0.60 to 1.15)
unknown risk	0.01	no	1.07	1.07	1.96	0.42 (0.08 to 0.80)	0.42 (0.08 to 0.80)

Supplementary Table S35. Clinical need increment with and without radiotherapy patients

stratum	model	clinical need added, primary cohort	clinical need added, prostatectomy without radiotherapy only
all men (pooled)	penalised logistic regression	3.01 (2.69 to 3.38)	1.66 (1.49 to 1.81)
all men (pooled)	LightGBM	3.18 (2.85 to 3.54)	1.66 (1.49 to 1.81)
low risk	penalised logistic regression	0.14 (0.08 to 0.20)	0.05 (-0.01 to 0.12)
low risk	LightGBM	0.02 (-0.09 to 0.13)	-0.15 (-0.28 to -0.02)
intermediate risk	penalised logistic regression	0.49 (0.41 to 0.56)	0.33 (0.21 to 0.44)
intermediate risk	LightGBM	0.55 (0.44 to 0.65)	0.06 (-0.06 to 0.17)

stratum	model	clinical need added, primary cohort	clinical need added, prostatectomy without radiotherapy only
high risk	penalised logistic regression	3.53 (3.16 to 3.86)	2.20 (1.96 to 2.42)
high risk	LightGBM	3.72 (3.34 to 4.06)	2.13 (1.88 to 2.32)
unknown risk	penalised logistic regression	0.65 (0.41 to 0.89)	0.91 (0.57 to 1.25)
unknown risk	LightGBM	0.59 (0.27 to 0.95)	0.62 (0.12 to 1.16)

Supplementary Tables S36 to S42. Secondary outcome: recorded curative treatment

No record can mean active surveillance, watchful waiting, hormone therapy only, refusal, or treatment the registry did not capture [20]. A lower percentage cannot be read as under-treatment.

Supplementary Table S36. By risk group

risk group	men	% with a recorded curative treatment
intermediate	178,250	75.4
high	174,390	81.1

Supplementary Table S37. By risk group and county rurality

risk group	rurality	men	% with a recorded curative treatment
intermediate	Metro, 1 million or more	104,530	76.0
intermediate	Metro, 250,000 to 1 million	39,410	75.3
intermediate	Metro, under 250,000	14,100	74.5
intermediate	Nonmetro, adjacent to metro	12,430	73.9
intermediate	Nonmetro, not adjacent to metro	7,700	72.3
intermediate	Unknown	80	53.2
high	Metro, 1 million or more	100,450	81.9
high	Metro, 250,000 to 1 million	38,520	82.1
high	Metro, under 250,000	13,730	78.9
high	Nonmetro, adjacent to metro	13,060	77.7
high	Nonmetro, not adjacent to metro	8,490	75.3
high	Unknown	150	63.9

Supplementary Table S38. By risk group and county income quartile

risk group	county income quartile	men	% with a recorded curative treatment
intermediate	Q1 (lowest)	14,140	72.7
intermediate	Q2	42,320	75.1
intermediate	Q3	61,910	75.7
intermediate	Q4 (highest)	59,870	76.0
intermediate	Unknown	20	36.4
high	Q1 (lowest)	14,280	73.7
high	Q2	41,500	79.6
high	Q3	62,280	82.0
high	Q4 (highest)	56,310	83.0
high	Unknown	30	51.6

Supplementary Table S39. By risk group and marital status

risk group	marital status	men	% with a recorded curative treatment
intermediate	Married (including common law)	118,930	79.0
intermediate	Single (never married)	20,560	72.8
intermediate	Divorced	12,520	74.5
intermediate	Separated	1,550	71.4
intermediate	Widowed	6,020	71.0
intermediate	Unmarried or Domestic Partner	790	74.6
intermediate	Unknown	17,890	56.8
high	Married (including common law)	115,360	86.2
high	Single (never married)	20,970	79.2
high	Divorced	12,170	80.7
high	Separated	1,560	78.2
high	Widowed	7,450	64.7

risk group	marital status	men	% with a recorded curative treatment
high	Unmarried or Domestic Partner	730	86.5
high	Unknown	16,140	54.7

Supplementary Table S40. Value added at each step

stratum	model	skill at step 2	clinical need added	social position added	social position as % of step 2 skill	AUC at step 2	calibration slope at step 2
intermediate and high risk (pooled)	penalised logistic regression	17.18	15.22 (14.85 to 15.65)	1.96 (1.72 to 2.31)	11	0.771	1.00
intermediate and high risk (pooled)	LightGBM	20.49	18.32 (17.92 to 18.77)	2.16 (1.92 to 2.54)	11	0.791	1.02
intermediate risk	penalised logistic regression	8.79	6.81 (6.38 to 7.21)	1.97 (1.65 to 2.42)	22	0.692	1.00
intermediate risk	LightGBM	10.14	7.83 (7.42 to 8.20)	2.31 (1.99 to 2.82)	23	0.703	1.02
high risk	penalised logistic regression	29.29	27.25 (26.63 to 27.96)	2.04 (1.84 to 2.33)	7	0.854	1.00
high risk	LightGBM	31.76	29.54 (28.94 to 30.17)	2.22 (2.00 to 2.53)	7	0.865	1.01

Supplementary Table S41. Standardised contrasts

stratum	contrast	penalised logistic regression	LightGBM	agreement
intermediate and high risk (pooled)	all social features as observed minus reference profile	-4.3	-3.5	both models 3 points or more, same direction
intermediate and high risk (pooled)	area: Nonmetro, not adjacent to metro minus Metro, 1 million or more, each at its typical county income	-2.9	-1.4	both models under 3 points
intermediate and high risk (pooled)	marital status: Single (never married) minus Married (including common law)	-6.5	-5.9	both models 3 points or more, same direction
intermediate risk	all social features as observed minus reference profile	-4.2	-3.2	both models 3 points or more, same direction
intermediate risk	area: Nonmetro, not adjacent to metro minus Metro, 1 million or more, each at its typical county income	-2.2	-0.3	both models under 3 points
intermediate risk	marital status: Single (never married) minus Married (including common law)	-7.2	-6.4	both models 3 points or more, same direction
high risk	all social features as observed minus reference profile	-4.3	-3.8	both models 3 points or more, same direction
high risk	area: Nonmetro, not adjacent to metro minus Metro, 1 million or more, each at its typical county income	-3.3	-2.5	same direction, only one model 3 points or more
high risk	marital status: Single (never married) minus Married (including common law)	-5.8	-5.1	both models 3 points or more, same direction

Supplementary Table S42. Standardised percentages by profile, intermediate and high risk pooled

profile	penalised logistic regression	LightGBM
as observed	78.2	78.2
reference profile	82.5	81.7
area: Metro, 1 million or more, typical county income (\$90,000 - \$94,999)	82.1	81.5
area: Metro, 250,000 to 1 million, typical county income (\$80,000 - \$84,999)	81.5	81.6
area: Metro, under 250,000, typical county income (\$65,000 - \$69,999)	80.7	81.1
area: Nonmetro, adjacent to metro, typical county income (\$55,000 - \$59,999)	79.7	80.1
area: Nonmetro, not adjacent to metro, typical county income (\$55,000 - \$59,999)	79.2	80.1
marital status: Married (including common law)	82.5	81.7
marital status: Single (never married)	76.0	75.9
marital status: Divorced	78.1	77.1
marital status: Separated	75.5	75.9
marital status: Widowed	78.4	78.3
marital status: Unmarried or Domestic Partner	79.8	77.4
marital status: Unknown	65.0	66.9

Supplementary Table S43. Protocol amendment log

d at e	amendment	reason
2 0 2 6- 0 9- 1 3	Timeliness made the primary question; treatment receipt secondary; survival reduced to a methods box	Strongest data, closest match to the intended audience, and time available
2 0 2 6- 0 9- 1 3	Analysis built on the current export without race, Type of Reporting Source or histology	Re-export not available before the application deadline; death-certificate proxy and stated limitations used instead
2 0 2 6- 0 9- 1 3	Modality moved after social position; ordered steps replace order-invariant decomposition; binary outcome only; cohort limited to 2010 to 2022; 0-day intervals excluded; cluster bootstrap over rurality by income cells	Independent design review: modality partly carries social effects; every treated man has the event, so survival framing does not apply; follow-up truncation; county-level exposures
2 0 2 6- 0 9- 1 3	Selection statement corrected: no age filter was applied at export	Session file inspection (reports/phase1.md)
2 0 2 6- 0 9- 1 3	Title changed to name the machine learning approach (version 1.1)	Clarity about methods; no change to the cohort, definitions, estimand or analyses
2 0 2 6- 0 9- 1 3	Wording clarified: no Australian data are analysed; Tasmanian findings are cited as background only; Australian registry variables are candidate equivalents to be confirmed (version 1.2)	Avoid implying a cross-country comparison or unverified registry contents; no change to the cohort, definitions, estimand or analyses
2 0 2 6- 0 9- 1 3	Survival methods box (A10) now prefers a time-dependent treatment exposure, with the 12-month landmark as a sensitivity analysis only; citations added for Leong 2025, Usman 2026 and Zheng 2023; references section added (version 1.3)	Simulation evidence that landmark methods only partly remove immortal time bias (Zheng et al. 2023). A10 is not estimated, so no result changes
2 0 2 6- 0 9- 1 3	Citations checked against full texts where available: Tasmanian findings restated with the published medians, mean differences and confidence intervals; the radiotherapy exclusion in Foley et al. 2025 stated; A11 updated with the area measures and data gaps reported in published PCOR-TAS analyses; references note records which papers were read in full (version 1.4)	Earlier wording was based on abstracts and was less precise; no change to the cohort, definitions, estimand or analyses
2 0 2 6- 0 9- 1 3	Modelling implementation details specified before any full-cohort model was run: hyperparameter grids, probability clipping, tuning procedure (once per model and stratum on the full-feature step, reused across steps, not nested), and the numeric definition of the top income tertile in the reference profile (version 1.5)	The protocol previously described these only in general terms; no change to the cohort, definitions, estimand or analyses
2 0 2 6- 0 9- 1 3	Deviation in A5: excess waiting days beyond 90 per 1,000 men are reported as crude observed values by rurality and county income quartile, not standardised, because no model for the number of days was built. Standardised contrasts are reported for the percentage delayed only, with intervals that hold the fitted model fixed (version 1.6)	Time available before the write-up; the primary estimand and the percentage-delayed contrasts are unaffected

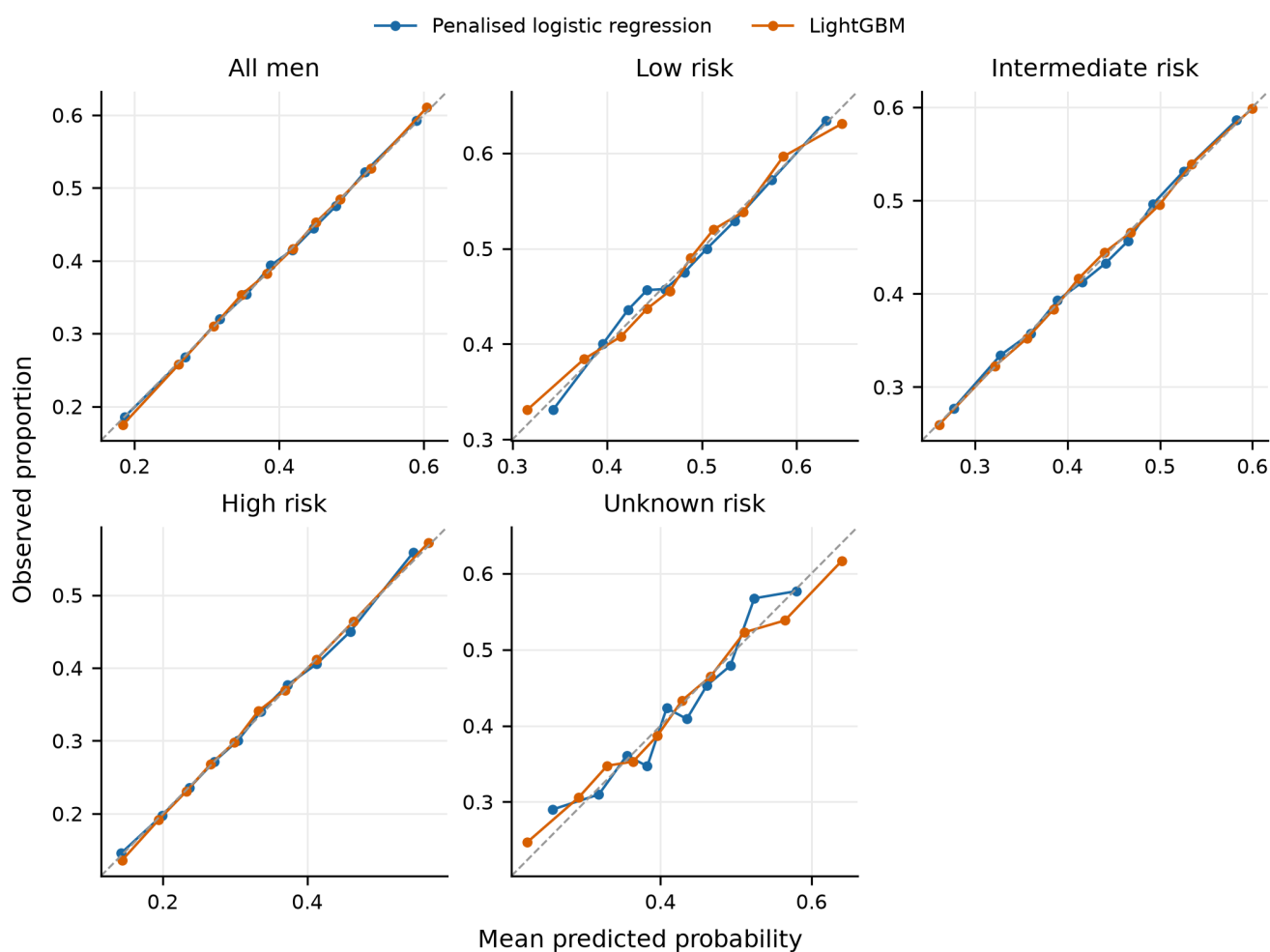
d at e	amendment	reason
2 0 2 6- 0 9- 1 3	A5 revised after a development run (5 bootstrap resamples). One-at-a-time rurality and county income profiles are no longer interpreted. The two variables are strongly correlated, so holding one fixed creates combinations rarely observed (for example remote counties in the top income tertile), and the two model types gave conflicting estimates for them. Joint area profiles (each rurality level at the median county income band of men living there) are reported instead, with the one-at-a-time results kept in a supplementary table. Bootstrap intervals that hold the fitted model fixed are not reported for standardised contrasts, because they ignore model-fitting uncertainty; a contrast is described as meaningful only when both model types agree on 3 percentage points or more in the same direction (version 1.7)	Prompted by the model disagreement seen in the development run; the rurality and income collinearity had been noted in Phase 3. The primary estimand (A2 to A4) is unaffected
2 0 2 6- 0 9- 1 3	A8 and A9 implementation specified before either was run (version 1.8). A8: each scenario rebuilds the cohort with exactly one change from the primary definition; a cohort option excluding men diagnosed in 2020 was added. For each scenario the report gives the number of men and percentage delayed by risk group, the social position increment (step 1 to 2) with cluster bootstrap intervals for both model types in every stratum, and the standardised area and marital status contrasts judged by the agreement rule of amendment 1.7. Hyperparameters are reused from the primary tuning for the same model and stratum; step 3 and leave-one-variable-out refits are not repeated. One model sensitivity analysis is added: penalised logistic regression re-tuned on a wider C grid (0.0001 to 10) in the primary cohort, with steps 0 to 2 refitted. A9: the outcome is a record of radical prostatectomy or radiotherapy in the first course of treatment. Men without such a record include men on active surveillance or watchful waiting, men treated with hormone therapy only, men who refused, and men whose treatment was not captured, so the outcome is described as a recorded curative treatment, not as treatment itself. Strata are intermediate risk, high risk, and both groups pooled. LightGBM is fitted alongside penalised logistic regression so that the agreement rule of amendment 1.7 can be applied. Hyperparameters are tuned once per model and stratum on step 2, because modality (step 3) is part of the outcome. Crude percentages with a recorded curative treatment by rurality, county income quartile and marital status use the same disclosure rules as A1	The protocol listed A8 and A9 without implementation detail. The wider C grid was added after Phase 4 showed C = 0.01 chosen at the edge of the pre-specified grid in every stratum, and is labelled as a post hoc check. No change to the primary cohort, estimand or reported Phase 4 results
2 0 2 6- 0 9- 1 3	Revision analyses specified after an internal multi-reviewer review of the draft preprint and before running them (version 1.9). All are labelled post-review. (a) Per-step tuning: hyperparameters are tuned separately at steps 0, 1 and 2 for each model and stratum, using the section 7 procedure, and the social position increment is re-estimated. (b) Cross-fitting variability: under per-step tuning, steps 0 to 2 are cross-fitted with 10 fold-assignment seeds per model and stratum; the mean, minimum and maximum increment across seeds are reported with the cell-bootstrap interval for the first seed. Tuning is not repeated per seed and no refitting bootstrap is run, for computing time; both are stated as limitations. (c) Clustering: the number of bootstrap cells and the cell size distribution are reported per stratum, and intervals are recomputed with county income band alone as a coarser clustering. (d) Stage-free clinical block: steps 0 to 2 are refitted with summary stage removed from the clinical block (risk strata unchanged). (e) Year specification: logistic regression is refitted with year of diagnosis as categories. (f) Year-aware concentration index: the Erreygers index is estimated by diagnosis period (2010 to 2014, 2015 to 2019, 2020 to 2022), with the county income quartile distribution by period. (g) Long intervals: the share of intervals above 365 days and the share top-coded are reported by risk group and by rurality. (h) The one-at-a-time rurality and income profiles computed under amendment 1.7 are published as a supplementary table, as that amendment stated	The review (paper/review/round1/synthesis.m d) found that headline intervals omit model-fitting variability, that reusing step 3 hyperparameters may weaken the LightGBM clinical model in small strata, that the concentration index is pooled over years, and that the promised one-at-a-time table was missing. These analyses test the robustness of reported results. The primary estimand is unchanged, and all results from versions 1.0 to 1.8 remain reported

Supplementary Table S44. Protocol versions by commit

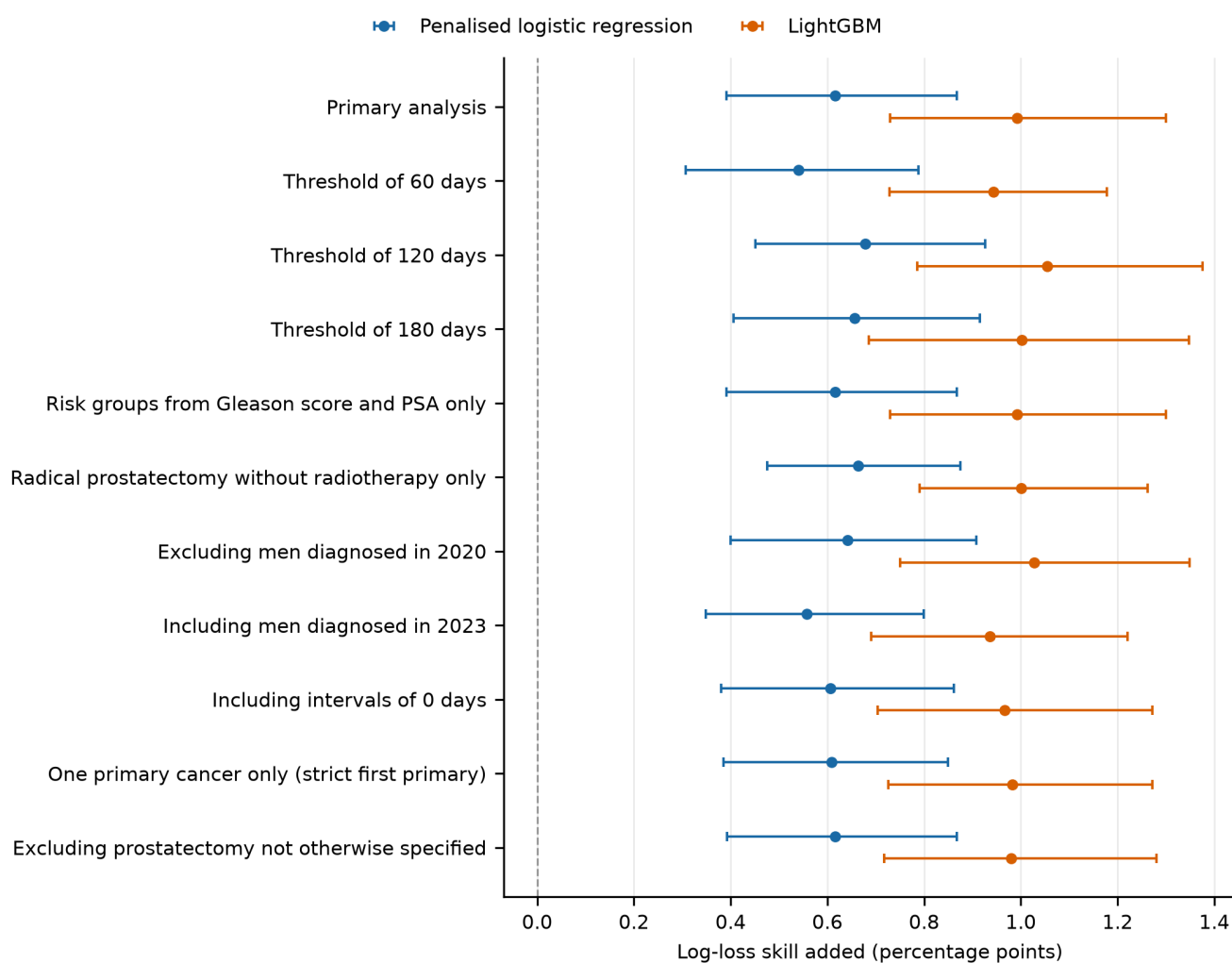
Commit times are recorded by the author's computer; the repository was first made public on 13 September 2026, after the analyses.

commit	commit time (author's computer)	protocol version	commit message
5954e76	2026-09-13 16:15:01 +0530	1.0	Add study protocol v1.0 and analysis definitions (2026-09-13)
5eb2280	2026-09-13 16:56:24 +0530	1.3	Add inventory, literature search, cohort pipeline and protocol v1.3
66a99f1	2026-09-13 16:59:55 +0530	1.4	Protocol v1.4: check citations against full texts
52982e0	2026-09-13 17:50:31 +0530	1.7	Phase 4: descriptive results, stepwise models, standardisation, equity and selection (A1 to A7)
d9b3851	2026-09-13 18:40:30 +0530	1.8	Phase 4: sensitivity analyses (A8) and recorded curative treatment (A9)
c1b6fd7	2026-09-13 21:24:50 +0530	1.9	Revision 1 after internal review: protocol amendment 1.9, robustness analyses, revised preprint

Supplementary figures



Supplementary Figure S1. Calibration of out-of-fold predicted probabilities at step 2 (clinical need plus area and marital characteristics), in 10 equal-count bins, by risk group and model type. The dashed line marks perfect calibration.



Supplementary Figure S2. Log-loss skill added by the area and marital block under each sensitivity scenario, all men, with 95% cluster bootstrap intervals. Each scenario changes one setting from the primary analysis. The threshold scenarios use a different outcome, so their values are not directly comparable with the others.

Supplementary Table S45. STROBE and RECORD checklist

Item wording is abbreviated from the RECORD statement checklist, which includes the STROBE items [13, 14]. Locations refer to sections of the main text unless marked S (supplement).

Item	Topic	Where reported
1a, 1b	Design in title or abstract; informative abstract	Abstract (Methods names a retrospective cohort study)
RECORD 1.1	Type of data and database named	Title and Abstract (SEER registry data)
RECORD 1.2	Region and time frame	Title and Abstract (US, 2010 to 2022)
RECORD 1.3	Linkage stated	Methods, Design: county attributes linked by SEER; no person-level linkage
2	Background and rationale	Introduction
3	Objectives	Introduction, final paragraph
4	Study design	Methods, Design
5	Setting, locations and dates	Methods, Design and Cohort
6a	Eligibility and selection	Methods, Cohort; Figure 1
6b	Matching	Not applicable
RECORD 6.1	Codes and algorithms for population selection	Methods, Cohort; config/analysis.yaml in the repository
RECORD 6.2	Validation of codes	Methods, Secondary outcome [20]; no validation study of the cohort algorithm
RECORD 6.3	Linkage flow diagram	Not applicable (no person-level linkage)
7	Outcomes, exposures, predictors defined	Methods, Outcome, Risk groups and Ordered feature steps
RECORD 7.1	Complete list of codes	config/analysis.yaml in the repository
8	Data sources and measurement	Methods, Design, Outcome and Risk groups
9	Efforts to address bias	Methods, Men without a recorded interval and Sensitivity analyses; Discussion, Strengths and limitations
10	Study size	Methods, Cohort (all eligible men, no sample size calculation); Figure 1
11	Quantitative variables	Methods, Ordered feature steps
12a	Statistical methods	Methods, Models and validation, Standardised percentages and Income concentration index
12b	Subgroups and interactions	Methods, Models and validation (risk-group strata; clinical interaction terms)
12c	Missing data	Methods, Ordered feature steps, Models and validation, and Men without a recorded interval
12d	Loss to follow-up	Not applicable (outcome is the recorded interval; top-coding described in Methods, Outcome)
12e	Sensitivity analyses	Methods, Sensitivity analyses and post-review robustness analyses
RECORD 12.1	Access to the database population	Methods, Design (full case listing of the selected site and years)
RECORD 12.2	Data cleaning	Methods, Cohort (explicit label mapping)
RECORD 12.3	Linkage methods	Not applicable (no person-level linkage)
13a to 13c	Numbers at each stage, reasons, flow diagram	Results, Cohort; Figure 1
RECORD 13.1	Selection of included persons	Methods, Cohort; Figure 1
14a, 14b	Characteristics and missing data	Table 1 (unknown categories shown)
14c	Follow-up time	Not applicable
15	Outcome data	Results, Waiting beyond 90 days; Tables S2 to S6
16a	Estimates and precision	Table 2 (intervals); Table 3 (no intervals, stated)
16b	Category boundaries	Methods, Risk groups; Table 1 note (income quartiles)
16c	Absolute measures	Results, Standardised percentages and excess days
17	Other analyses	Results, Sensitivity analyses and post-review robustness analyses; Tables S11 to S15, S29 to S35
18	Key results	Discussion, Principal findings
19	Limitations	Discussion, Strengths and limitations
RECORD 19.1	Implications of routinely collected data	Discussion, Strengths and limitations; What the direction can and cannot show
20	Interpretation	Discussion
21	Generalisability	Discussion, Australian context and Strengths and limitations
22	Funding	Declarations
RECORD 22.1	Access to protocol, data and code	Methods, Design; Declarations

Supplementary Table S46. TRIPOD+AI checklist

Item wording is abbreviated from the TRIPOD+AI checklist [15]. Development and internal evaluation use the same data through cross-fitting; the models are measurement tools and are not proposed for clinical use.

Item	Topic	Where reported
1	Title	Title (prediction, population and outcome named)
2	Abstract	Abstract
3a	Healthcare context and rationale	Introduction
3b	Target population and intended purpose	Introduction, final paragraph; Methods, Models and validation (not intended for clinical use)
3c	Known health inequalities	Introduction, second paragraph
4	Objectives	Introduction, final paragraph
5a	Data sources	Methods, Design
5b	Dates of data	Methods, Design and Cohort
6a	Setting	Methods, Design (population-based registries)
6b	Eligibility	Methods, Cohort
6c	Treatments	Methods, Cohort and Ordered feature steps (treatment type at step 3)
7	Data preparation	Methods, Cohort (label mapping)
8a	Outcome definition	Methods, Outcome
8b, 8c	Outcome assessors and blinding	Not applicable (registry-abstracted outcome)
9a	Choice of predictors	Methods, Ordered feature steps (pre-specified in the protocol)
9b	Predictor definitions	Methods, Risk groups and Ordered feature steps; config/analysis.yaml
9c	Predictor assessors	Not applicable (registry-coded predictors)
10	Sample size	Methods, Cohort (all eligible men)
11	Missing data	Methods, Ordered feature steps, Models and validation, and Men without a recorded interval
12a	Data use and partitioning	Methods, Models and validation (5-fold cross-fitting)
12b	Predictor handling	Methods, Ordered feature steps and Models and validation
12c	Model type, building steps, tuning, internal validation	Methods, Models and validation
12d	Heterogeneity across clusters	Methods, Models and validation (cluster bootstrap only; heterogeneity across clusters not quantified)
12e	Performance measures	Methods, Models and validation; Supplementary Figure S1
12f	Model updating	Not applicable
12g	How predictions were calculated	Code in the repository
13	Class imbalance	Methods, Models and validation (none used)
14	Fairness	Not addressed as a model property; differences by area and marital status are the study's measured quantity
15	Model output	Predicted probabilities; no classification threshold
16	Training versus evaluation differences	Not applicable (cross-fitting within the same data)
17	Ethical approval	Methods, Design; Declarations
18a to 18f	Funding, conflicts, protocol, registration, data and code sharing	Methods, Design; Declarations
19	Patient and public involvement	Methods, Design; Declarations (none)
20a	Participant flow and outcome counts	Results, Cohort and Waiting beyond 90 days; Figure 1
20b	Participant characteristics	Table 1
20c	Comparison with development data	Not applicable
21	Participants and events per analysis	Table 1 (men waiting more than 90 days); Table 2 and Supplementary Table S2 (men and percentage delayed per stratum); Methods, Models and validation (tuning on up to 50,000 men)
22	Full model specification	Not released as a clinical model; code that refits every model is in the repository
23a	Performance with confidence intervals, including subgroups	Table 2 and Table S8 by risk group (skill increments with intervals; AUC and calibration without intervals); performance by area or marital subgroup not reported
23b	Heterogeneity across clusters	Not examined
24	Model updating results	Not applicable
25	Interpretation	Discussion
26	Limitations	Discussion, Strengths and limitations
27a, 27b	Input data handling and user interaction in implementation	Not applicable (no implementation proposed)
27c	Next steps	Discussion, Australian context; Conclusions